

Assessing Mediterranean Swordfish Using the Assessment for All (a4a) Model

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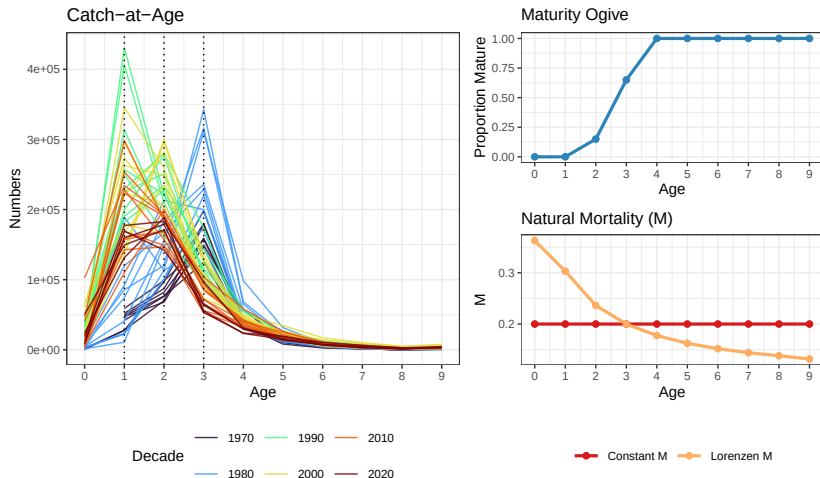
SCRS/2026/170

Background

- **Context:** Following the update of the 2020 baseline $\alpha_4\alpha$ assessment through 2024, structural enhancements were investigated to address historical data coverage and biological realism. The new catch at age matrix support using a plus group at age 9.
- **Key Objectives:**
 - **Temporal Scope:** Evaluate sensitivity to the time-series start year by comparing the truncated period (**1985–2024**) against the extended historical series (**1978–2024**).
 - **Natural Mortality (M):** Assess the impact of assuming a **Constant M** versus an age-dependent **Lorenzen M** vector.
 - **Fishing Mortality Dynamics:** Incorporate a tensor product interaction ($\text{ti}(\text{year}, \text{age})$) to capture time-varying selectivity across years and age classes.

Input Data and Assumptions

Catch at age 0-9 with plus group set at 9



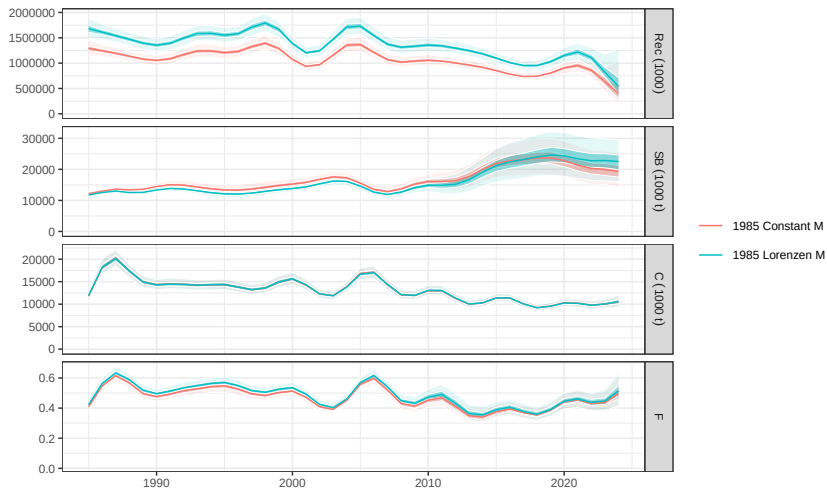
Summary of Candidate & Sensitivity Runs

Run	Period	M Assumption	Fbar Range	Indices
Run 1	1985–2024	Constant M	Ages 1–4	5 indices
Run 2	1985–2024	Lorenzen M	Ages 1–4	5 indices
Sensitivity 1	1978–2024	Constant M	Ages 1–4	5 indices
Sensitivity 2	1985–2024	Constant M	Ages 1–4	5 indices + SI_GN + LI_SUR

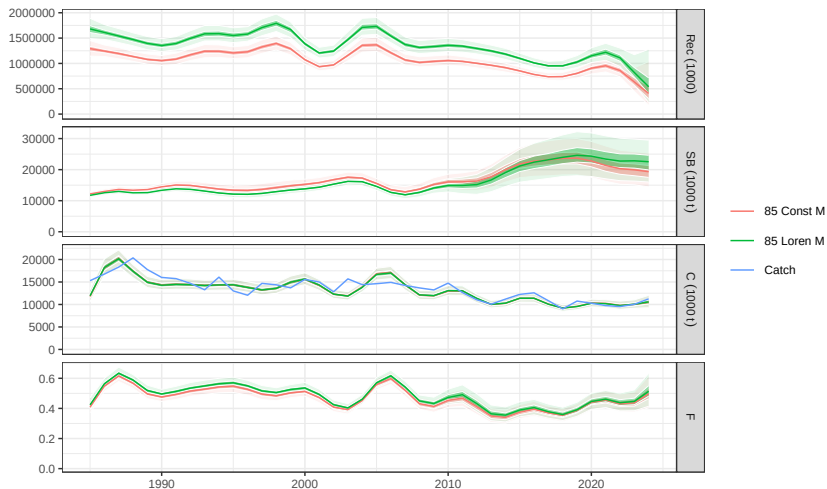
Model Specifications & Sub-models

Component	Specification
F sub-model (1985 runs)	$\sim s(\text{year}, k = 20) + s(\text{age}, k = 8) + \text{ti}(\text{year}, \text{age}, k = c(6, 5))$
SR sub-model	$\sim s(\text{year}, k = 20)$
Q sub-model (GR, SP, SI)	~ 1 (Constant catchability)
Q sub-model (LI, MO)	$\sim s(\text{year}, k = 3)$ (Time-varying catchability)

Trajectory Comparison: 1985 Series (Constant vs. Lorenzen M)



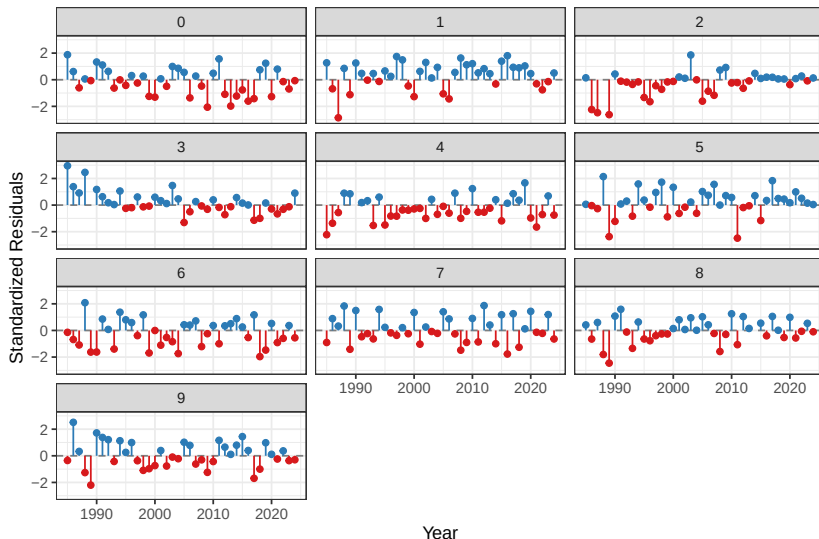
Candidate Runs vs. Observed Catch



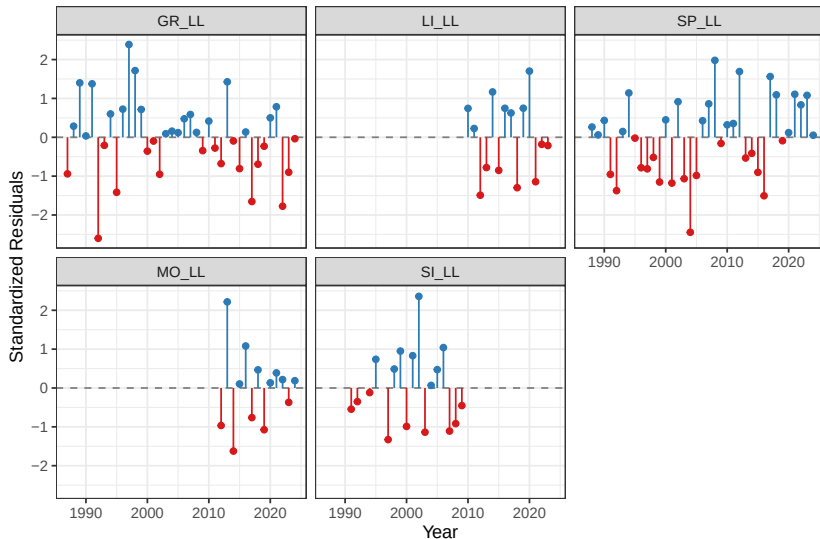
Section 1

Diagnostics - Constant M

Catch-at-Age Standardized Residuals

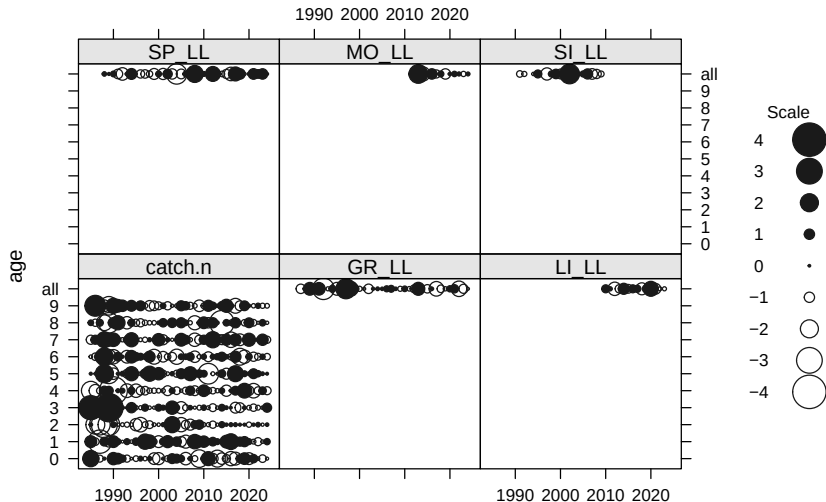


Standardized CPUE Index Residuals



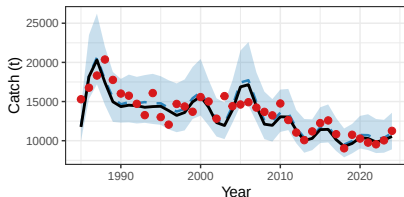
Catch-at-Age Bubble Plot

log residuals of catch and abundance indices

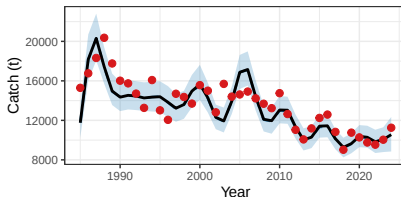


Aggregated catch diagnostics

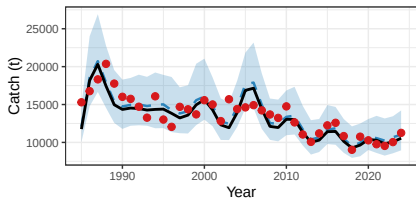
Observation Error (oe)



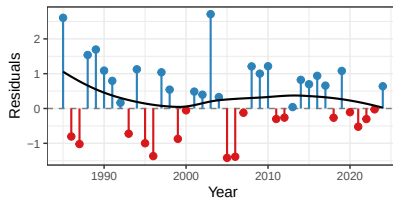
Estimation Error (ee)



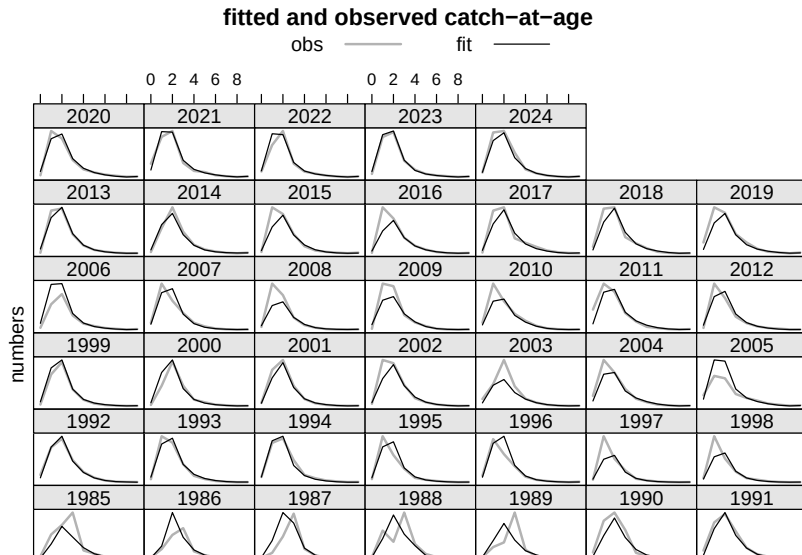
Prediction Error (oe)



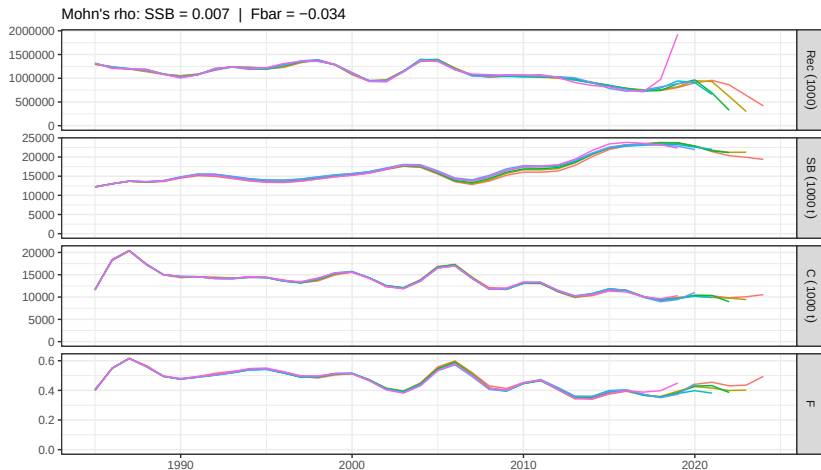
Standardized Catch Residuals (resstd)



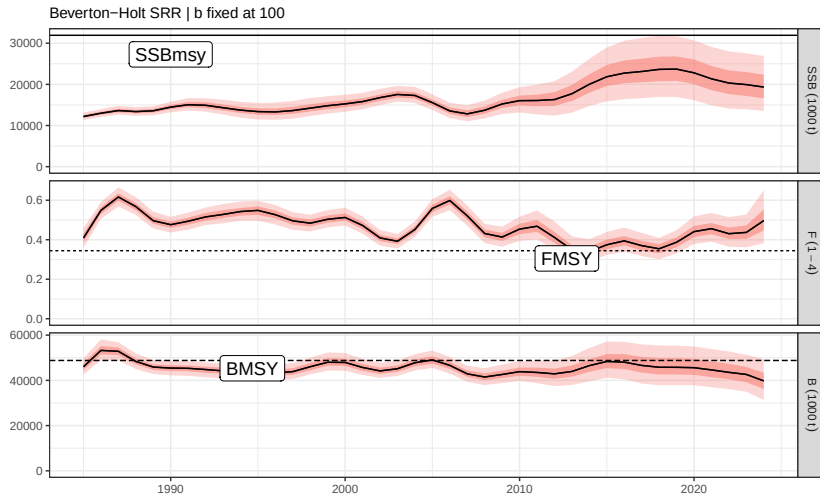
Observed vs. Fitted Stock - Catch (85 Const M)



Retrospective Analysis (Peels = 5)



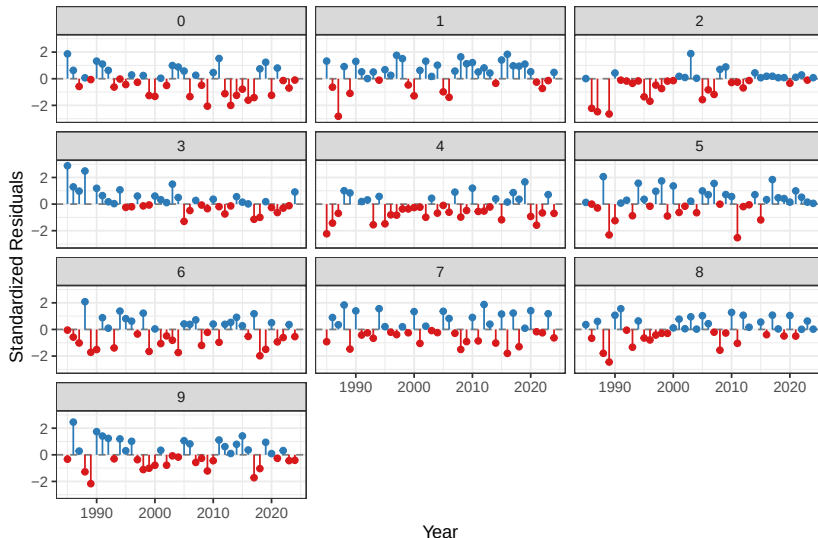
Stock status



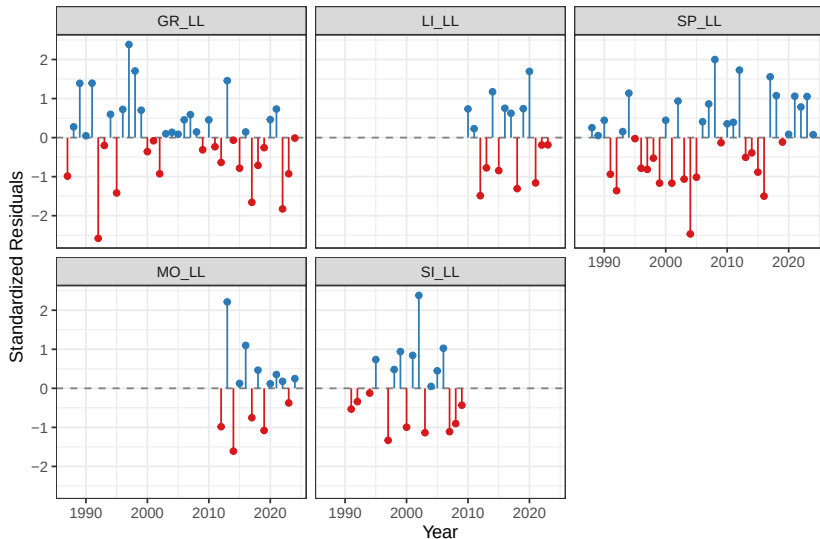
Section 2

Diagnostics - Lorenzen M

Catch-at-Age Standardized Residuals

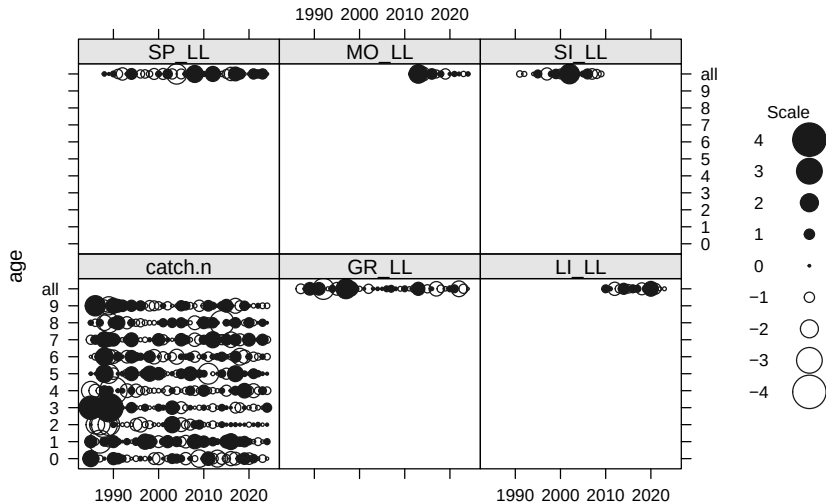


Standardized CPUE Index Residuals



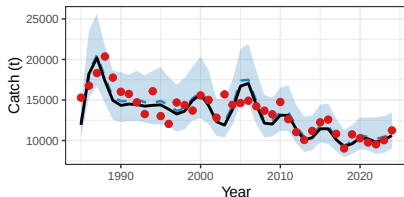
Catch-at-Age Bubble Plot

log residuals of catch and abundance indices

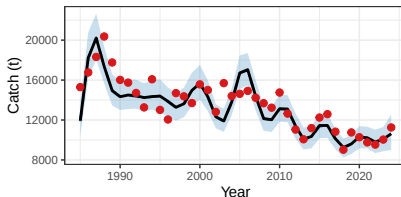


Aggregated catch diagnostics

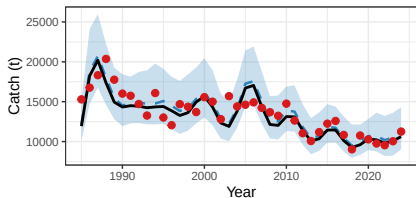
Observation Error (oe)



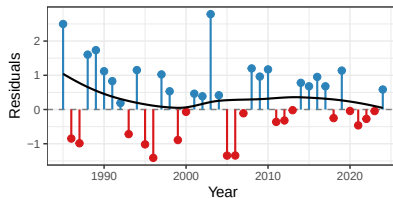
Estimation Error (ee)



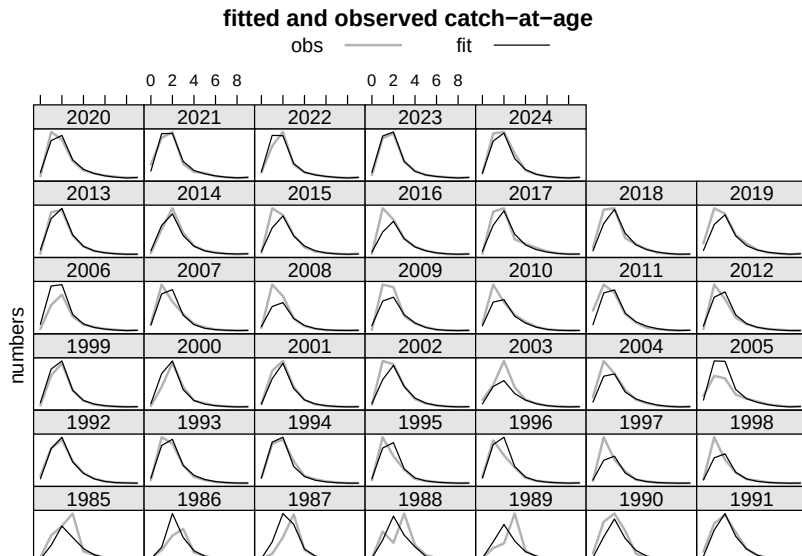
Prediction Error (oee)



Standardized Catch Residuals (resstd)

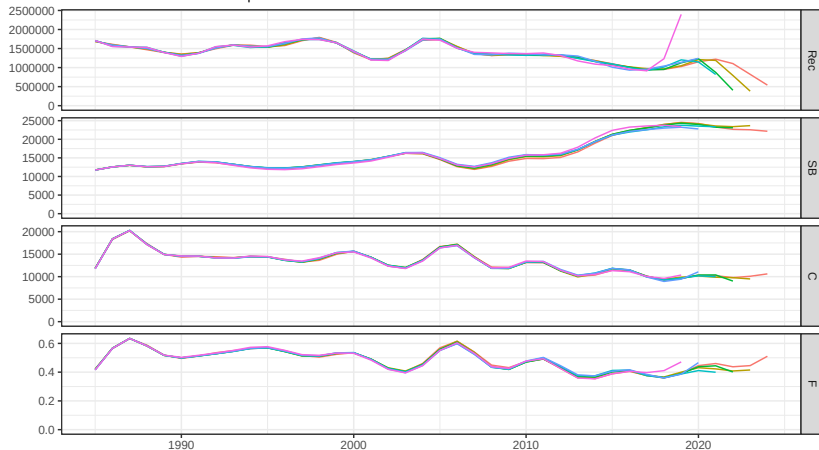


Observed vs. Fitted Stock - Catch

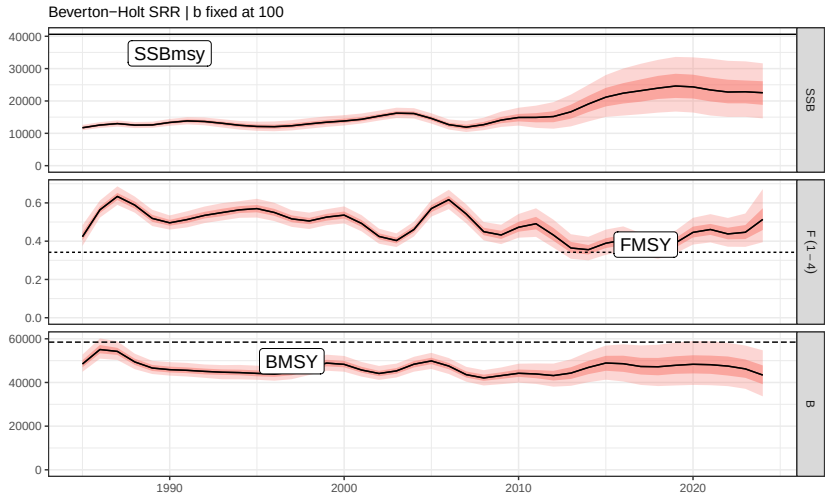


Retrospective Analysis (Peels = 5)

Mohn's rho: SSB = -0.007 | Fbar = -0.006



Stock status

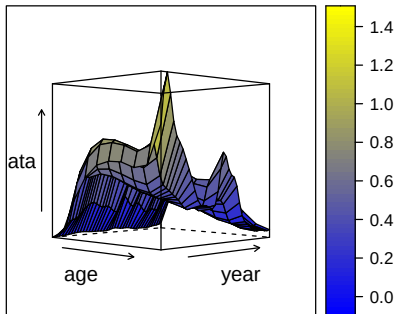


Section 3

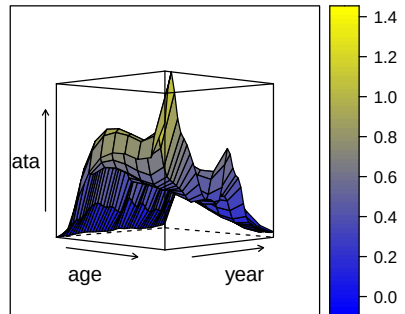
Candidate runs comparison

Fishing mortality 3D plots

Fishing mortality



Fishing mortality



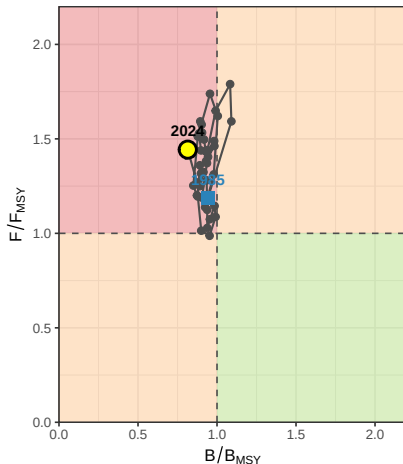
Stock Status & Reference Points

Candidate Runs Comparison

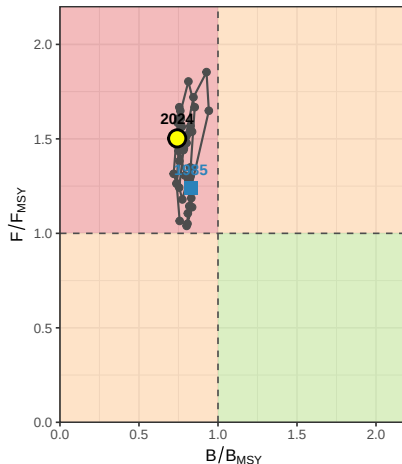
Metric	1985 Constant M	1985 Lorenzen M
FMSY	0.344	0.344
BMSY (t)	48803.874	48803.874
F2024	0.495	0.510
B2024 (t)	39062.288	42468.375
F2024 / FMSY	1.436	1.493
B2024 / BMSY	0.800	0.726

Stock Status Trajectories (F/F_{MSY} vs. B/B_{MSY})

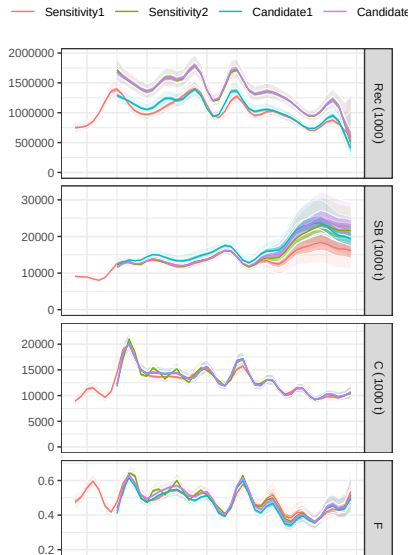
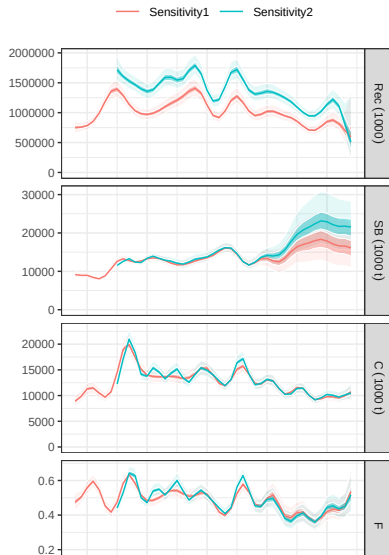
Kobe Plot: 1985 Constant M



Kobe Plot: 1985 Lorenzen M



Sensitivity runs



Sensitivity runs - Stock status

